

Practice Problems Sheet X

(Generating Functions)

Q1 Find a closed form generating function for sequences:

(a) $a_n = \binom{n+4}{n}$, for $n = 0, 1, 2, \dots$

(b) $a_n = \binom{10}{n+1}$, for $n = 0, 1, 2, \dots$

Q2 Derive a closed form formula for the sequence determined by generating function:

(a) $(x^4/(1-x^4)) - x^3 - x^2 - x - 1$

(b) $x^2/(1-x)^2$

Q3 Give a combinatorial interpretation of the coefficient of x^6 in the expansion $(1+x+x^2+x^3+\dots)^n$. Use this interpretation to find this number.

Q4

(a) What is the generating function for $\{a_k\}$, where a_k is the number of solutions of $x_1+x_2+x_3+x_4=k$. Here x_1, x_2, x_3 and x_4 are integer variables with constraints $x_1 \geq 3, 1 \leq x_2 \leq 5, 0 \leq x_3 \leq 4$ and $x_4 \geq 1$?

(b) Use your answer to part (a) to find a_7 .

Q5 If $G(x)$ is the generating function for $\{a_k\}$, what is the closed form generating function for each of these sequences?

(a) $0, 0, 0, a_3, a_4, \dots$

(b) $a_0, 0, a_1, 0, a_2, 0, \dots$

(c) $0, a_0, a_1/2, a_2/3, a_3/4, \dots$

(d) $a_0^2, 2a_0a_1, 2a_0a_2 + a_1^2, 2a_0a_3 + 2a_1a_2, \dots$

Q6 Using Generating functions solve the recurrence relation $a_k = 2a_{k-1} + 3a_{k-2} + 4^k + 6$ with initial conditions $a_0 = 20$ and $a_1 = 60$.

Q7 Observe that exponential generating function of sequence $1, 1, 1, \dots$ is the same as ordinary generating function of sequence $\left\{\frac{1}{n!}\right\}$.

Q8 Find a closed form for the exponential generating function for the sequence $\{a_n\}$, where

(a) $a_n = n + 1$

(b) $a_n = \frac{1}{n+1}$

(c) $a_n = \frac{1}{(n+1)(n+2)}$

Q9 Find the sequence for each of these functions as its exponential generating function.

(a) $f(x) = e^{-2x} - \frac{1}{1-x}$

(b) $f(x) = e^{x^2}$

—————End—————